$$P + Qy'' = 0$$

!

a 2nd Order Differential Equation

<u>Definition</u>

We say

$$P(x,y) + Q(x,y) \cdot y' = 0$$

is <u>exact</u> if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

in which case there is a $\Psi(x,y)$ such that $\frac{\partial \Psi}{\partial x} = P$ and $\frac{\partial \Psi}{\partial y} = Q$.

Further $\Psi(x,y) = C$ is an implicit solution to this differential equation.

$\boxed{\text{example}}$ $$\frac{x}{\sqrt{x^2+y^2}} + \frac{y}{\sqrt{x^2+y^2}} \cdot y' = 0$$

first <u>check exact</u> : $\frac{\partial P}{\partial y} = \frac{-xy}{(x^2+y^2)^{3/2}} = \frac{\partial Q}{\partial x}$ ✓

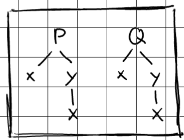
$\boxed{\text{solution}}$: $$\frac{\partial \Psi}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} \xrightarrow{u = x^2+y^2} \Psi(x,y) = \sqrt{x^2+y^2} + g(y)$$

$$\Big\downarrow \frac{\partial}{\partial y}$$

$$\frac{y}{\sqrt{x^2+y^2}} = \frac{y}{\sqrt{x^2+y^2}} + \underbrace{g'(y)}_{=0}$$

$$\boxed{\sqrt{x^2+y^2} = C}$$

towards exact 2^{nd} order D.E.



$$P(x,y) + Q(x,y)\,y' = 0$$

$\frac{d}{dx}$

$$\frac{d}{dx}P(x,y) + \frac{d}{dx}Q(x,y)\,y' = 0$$

$$\Rightarrow \frac{\partial P}{\partial x} + \frac{\partial P}{\partial y}y' + \left(\frac{\partial Q}{\partial x} + \frac{\partial Q}{\partial y}y'\right)\cdot y' + Q(x,y)\cdot y'' = 0$$

$$\boxed{\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}}$$

$$\square\, y'' + \square\,(y')^2 + \square\, y' + \square = 0$$

special case $\to$ $\frac{d}{dx}(\overbrace{A(x)}^{Q}y') + \frac{d}{dx}(\overbrace{B(x)y}^{P(x)y}) = 0 \quad \Rightarrow B(x) = A'(x)$

(new exactness cond.

$\boxed{\text{Definition}}$ $\leftarrow$ only for today

We say

$$p(x)y'' + q(x)y' + r(x)y = 0$$

is exact if it can be written

$$\frac{d}{dx}(A(x)y') + \frac{d}{dx}(B(x)y) = 0$$

example: $e^x y'' + (2x + e^x)y' + x^2 y = 0$

$$(e^x y'' + e^x y') + (2xy' + x^2 y) = 0$$

$$\frac{d}{dx}(e^x y') + \frac{d}{dx}(x^2 y) = 0$$

$$\Rightarrow \frac{d}{dx}\left(e^x y' + x^2 y\right) = 0 \quad \Rightarrow \quad e^x y' + x^2 y = C$$

$$\Rightarrow y' + x^2 e^{-x} + Ce^{-x}$$

$$\boxed{\alpha(x) = e^{(-x^2-2x-2)e^{-x}}}$$

$$y = \frac{1}{\alpha(x)}\left(C_0 + c\int \alpha(x)e^{-x}\,dx\right)$$